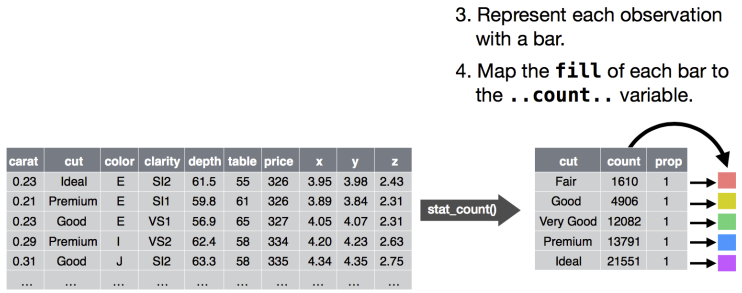
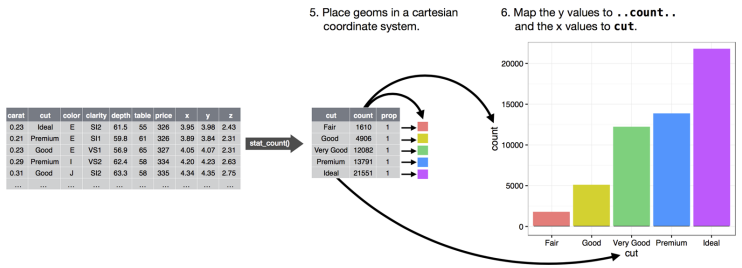
Next, you could choose a geometric object to represent each observation in the transformed data. You could then use the aesthetic properties of the geoms to represent variables in the data. You would map the values of each variable to the levels of an aesthetic:



You'd then select a coordinate system to place the geoms into. You'd use the location of the objects (which is itself an aesthetic property) to display the values of the x and y variables. At that point, you would have a complete graph, but you could further adjust the positions of the geoms within the coordinate system (a position adjustment) or split the graph into subplots (faceting). You could also extend the plot by adding one or more additional layers, where each additional layer uses a dataset, a geom, a set of mappings, a stat, and a position adjustment:



You could use this method to build *any* plot that you imagine. In other words, you can use the code template that you've learned in this chapter to build hundreds of thousands of unique plots.

CHAPTER 2

Workflow: Basics

You now have some experience running R code. I didn't give you many details, but you've obviously figured out the basics, or you would've thrown this book away in frustration! Frustration is natural when you start programming in R, because it is such a stickler for punctuation, and even one character out of place will cause it to complain. But while you should expect to be a little frustrated, take comfort in that it's both typical and temporary: it happens to everyone, and the only way to get over it is to keep trying.

Before we go any further, let's make sure you've got a solid foundation in running R code, and that you know about some of the most helpful RStudio features.

Coding Basics

Let's review some basics we've so far omitted in the interests of getting you plotting as quickly as possible. You can use R as a calculator:

```
1 / 200 * 30
#> [1] 0.15
(59 + 73 + 2) / 3
#> [1] 44.7
sin(pi / 2)
#> [1] 1
```

You can create new objects with `<-`:

```
x <- 3 * 4
```

All R statements where you create objects, *assignment* statements, have the same form:

```
object_name <- value
```

When reading that code say "object name gets value" in your head.

You will make lots of assignments and `<-` is a pain to type. Don't be lazy and use `=`: it will work, but it will cause confusion later. Instead, use RStudio's keyboard shortcut: Alt-– (the minus sign). Notice that RStudio automagically surrounds `<-` with spaces, which is a good code formatting practice. Code is miserable to read on a good day, so giveyoureyesabreak and use spaces.

What's in a Name?

Object names must start with a letter, and can only contain letters, numbers, `_`, and `.`. You want your object names to be descriptive, so you'll need a convention for multiple words. I recommend *snake_case* where you separate lowercase words with `_`:

```
i_use_snake_case
otherPeopleUseCamelCase
some.people.use.periods
And_aFew.People_RENOUNCEconvention
```

We'll come back to code style later, in Chapter 15.

You can inspect an object by typing its name:

```
x
#> [1] 12
```

Make another assignment:

```
this_is_a_really_long_name <- 2.5
```

To inspect this object, try out RStudio's completion facility: type "this," press Tab, add characters until you have a unique prefix, then press Return.

Oops, you made a mistake! `this_is_a_really_long_name` should have value 3.5 not 2.5. Use another keyboard shortcut to help you fix it. Type "this" then press Cmd/Ctrl-↑. That will list all the commands you've typed that start with those letters. Use the arrow keys to navigate, then press Enter to retype the command. Change 2.5 to 3.5 and rerun.

Make yet another assignment:

```
r_rocks <- 2 ^ 3
```

Let's try to inspect it:

```
r_rock
#> Error: object 'r_rock' not found
R_rocks
#> Error: object 'R_rocks' not found
```

There's an implied contract between you and R: it will do the tedious computation for you, but in return, you must be completely precise in your instructions. Typos matter. Case matters.

Calling Functions

R has a large collection of built-in functions that are called like this:

```
function_name(arg1 = val1, arg2 = val2, ...)
```

Let's try using `seq()`, which makes regular *seq*uences of numbers and, while we're at it, learn more helpful features of RStudio. Type `se` and hit Tab. A pop-up shows you possible completions. Specify `seq()` by typing more (a "q") to disambiguate, or by using ↑/↓ arrows to select. Notice the floating tooltip that pops up, reminding you of the function's arguments and purpose. If you want more help, press F1 to get all the details in the help tab in the lower-right pane.

Press Tab once more when you've selected the function you want. RStudio will add matching opening (`(`) and closing (`)`) parentheses for you. Type the arguments `1, 10` and hit Return:

```
seq(1, 10)
#>  [1]  1  2  3  4  5  6  7  8  9 10
```

Type this code and notice similar assistance help with the paired quotation marks:

```
x <- "hello world"
```

Quotation marks and parentheses must always come in a pair. RStudio does its best to help you, but it's still possible to mess up and end up with a mismatch. If this happens, R will show you the continuation character "+":

```
> x <- "hello
+
```

The + tells you that R is waiting for more input; it doesn't think you're done yet. Usually that means you've forgotten either a " or a). Either add the missing pair, or press Esc to abort the expression and try again.

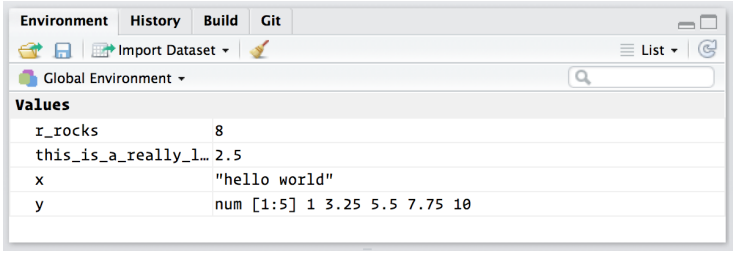
If you make an assignment, you don't get to see the value. You're then tempted to immediately double-check the result:

```
y <- seq(1, 10, length.out = 5)
y
#> [1]  1.00  3.25  5.50  7.75 10.00
```

This common action can be shortened by surrounding the assignment with parentheses, which causes assignment and "print to screen" to happen:

```
(y <- seq(1, 10, length.out = 5))
#> [1]  1.00  3.25  5.50  7.75 10.00
```

Now look at your environment in the upper-right pane:



Here you can see all of the objects that you've created.

Exercises

1. Why does this code not work?

   ```
   my_variable <- 10
   my_variable
   #> Error in eval(expr, envir, enclos):
   #> object 'my_variable' not found
   ```

   Look carefully! (This may seem like an exercise in pointlessness, but training your brain to notice even the tiniest difference will pay off when programming.)

2. Tweak each of the following R commands so that they run correctly:

```
library(tidyverse)

ggplot(dota = mpg) +
  geom_point(mapping = aes(x = displ, y = hwy))

fliter(mpg, cyl = 8)
filter(diamond, carat > 3)
```

3. Press Alt-Shift-K. What happens? How can you get to the same place using the menus?

CHAPTER 3

Data Transformation with dplyr

Introduction

Visualization is an important tool for insight generation, but it is rare that you get the data in exactly the right form you need. Often you'll need to create some new variables or summaries, or maybe you just want to rename the variables or reorder the observations in order to make the data a little easier to work with. You'll learn how to do all that (and more!) in this chapter, which will teach you how to transform your data using the **dplyr** package and a new dataset on flights departing New York City in 2013.

Prerequisites

In this chapter we're going to focus on how to use the **dplyr** package, another core member of the tidyverse. We'll illustrate the key ideas using data from the **nycflights13** package, and use **ggplot2** to help us understand the data.

```
library(nycflights13)
library(tidyverse)
```

Take careful note of the conflicts message that's printed when you load the tidyverse. It tells you that **dplyr** overwrites some functions in base R. If you want to use the base version of these functions after loading **dplyr**, you'll need to use their full names: `stats::filter()` and `stats::lag()`.

nycflights13

To explore the basic data manipulation verbs of **dplyr**, we'll use `nycflights13::flights`. This data frame contains all 336,776 flights that departed from New York City in 2013. The data comes from the US Bureau of Transportation Statistics, and is documented in `?flights`:

```
flights
#> # A tibble: 336,776 × 19
#>    year month   day dep_time sched_dep_time dep_delay
#>   <int> <int> <int>    <int>          <int>     <dbl>
#> 1  2013     1     1      517            515         2
#> 2  2013     1     1      533            529         4
#> 3  2013     1     1      542            540         2
#> 4  2013     1     1      544            545        -1
#> 5  2013     1     1      554            600        -6
#> 6  2013     1     1      554            558        -4
#> # ... with 336,776 more rows, and 13 more variables:
#> #   arr_time <int>, sched_arr_time <int>, arr_delay <dbl>,
#> #   carrier <chr>, flight <int>, tailnum <chr>, origin <chr>,
#> #   dest <chr>, air_time <dbl>, distance <dbl>, hour <dbl>,
#> #   minute <dbl>, time_hour <dttm>
```

You might notice that this data frame prints a little differently from other data frames you might have used in the past: it only shows the first few rows and all the columns that fit on one screen. (To see the whole dataset, you can run `View(flights)`, which will open the dataset in the RStudio viewer.) It prints differently because it's a *tibble*. Tibbles are data frames, but slightly tweaked to work better in the tidyverse. For now, you don't need to worry about the differences; we'll come back to tibbles in more detail in Part II.

You might also have noticed the row of three- (or four-) letter abbreviations under the column names. These describe the type of each variable:

- `int` stands for integers.
- `dbl` stands for doubles, or real numbers.
- `chr` stands for character vectors, or strings.
- `dttm` stands for date-times (a date + a time).

There are three other common types of variables that aren't used in this dataset but you'll encounter later in the book:

- `lgl` stands for logical, vectors that contain only `TRUE` or `FALSE`.
- `fctr` stands for factors, which R uses to represent categorical variables with fixed possible values.
- `date` stands for dates.

dplyr Basics

In this chapter you are going to learn the five key **dplyr** functions that allow you to solve the vast majority of your data-manipulation challenges:

- Pick observations by their values (`filter()`).
- Reorder the rows (`arrange()`).
- Pick variables by their names (`select()`).
- Create new variables with functions of existing variables (`mutate()`).
- Collapse many values down to a single summary (`summarize()`).

These can all be used in conjunction with `group_by()`, which changes the scope of each function from operating on the entire dataset to operating on it group-by-group. These six functions provide the verbs for a language of data manipulation.

All verbs work similarly:

1. The first argument is a data frame.
2. The subsequent arguments describe what to do with the data frame, using the variable names (without quotes).
3. The result is a new data frame.

Together these properties make it easy to chain together multiple simple steps to achieve a complex result. Let's dive in and see how these verbs work.

Filter Rows with filter()

`filter()` allows you to subset observations based on their values. The first argument is the name of the data frame. The second and

subsequent arguments are the expressions that filter the data frame. For example, we can select all flights on January 1st with:

```
filter(flights, month == 1, day == 1)
#> # A tibble: 842 × 19
#>    year month   day dep_time sched_dep_time dep_delay
#>   <int> <int> <int>    <int>          <int>     <dbl>
#> 1  2013     1     1      517            515         2
#> 2  2013     1     1      533            529         4
#> 3  2013     1     1      542            540         2
#> 4  2013     1     1      544            545        -1
#> 5  2013     1     1      554            600        -6
#> 6  2013     1     1      554            558        -4
#> # ... with 836 more rows, and 13 more variables:
#> #   arr_time <int>, sched_arr_time <int>, arr_delay <dbl>,
#> #   carrier <chr>, flight <int>, tailnum <chr>,origin <chr>,
#> #   dest <chr>, air_time <dbl>, distance <dbl>, hour <dbl>,
#> #   minute <dbl>, time_hour <dttm>
```

When you run that line of code, **dplyr** executes the filtering operation and returns a new data frame. **dplyr** functions never modify their inputs, so if you want to save the result, you'll need to use the assignment operator, `<-`:

```
jan1 <- filter(flights, month == 1, day == 1)
```

R either prints out the results, or saves them to a variable. If you want to do both, you can wrap the assignment in parentheses:

```
(dec25 <- filter(flights, month == 12, day == 25))
#> # A tibble: 719 × 19
#>    year month   day dep_time sched_dep_time dep_delay
#>   <int> <int> <int>    <int>          <int>     <dbl>
#> 1  2013    12    25      456            500        -4
#> 2  2013    12    25      524            515         9
#> 3  2013    12    25      542            540         2
#> 4  2013    12    25      546            550        -4
#> 5  2013    12    25      556            600        -4
#> 6  2013    12    25      557            600        -3
#> # ... with 713 more rows, and 13 more variables:
#> #   arr_time <int>, sched_arr_time <int>, arr_delay <dbl>,
#> #   carrier <chr>, flight <int>, tailnum <chr>,origin <chr>,
#> #   dest <chr>, air_time <dbl>, distance <dbl>, hour <dbl>,
#> #   minute <dbl>, time_hour <dttm>
```

Comparisons

To use filtering effectively, you have to know how to select the observations that you want using the comparison operators. R provides the standard suite: `>`, `>=`, `<`, `<=`, `!=` (not equal), and `==` (equal).

When you're starting out with R, the easiest mistake to make is to use = instead of == when testing for equality. When this happens you'll get an informative error:

```
filter(flights, month = 1)
#> Error: filter() takes unnamed arguments. Do you need `==`?
```

There's another common problem you might encounter when using ==: floating-point numbers. These results might surprise you!

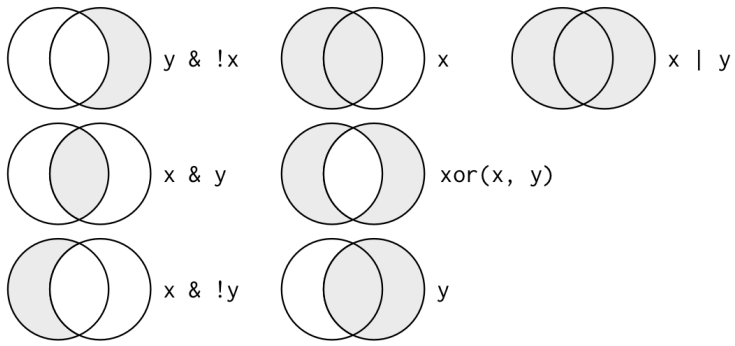
```
sqrt(2) ^ 2 == 2
#> [1] FALSE
1/49 * 49 == 1
#> [1] FALSE
```

Computers use finite precision arithmetic (they obviously can't store an infinite number of digits!) so remember that every number you see is an approximation. Instead of relying on ==, use `near()`:

```
near(sqrt(2) ^ 2,  2)
#> [1] TRUE
near(1 / 49 * 49, 1)
#> [1] TRUE
```

Logical Operators

Multiple arguments to `filter()` are combined with "and": every expression must be true in order for a row to be included in the output. For other types of combinations, you'll need to use Boolean operators yourself: `&` is "and," `|` is "or," and `!` is "not." The following figure shows the complete set of Boolean operations.



The following code finds all flights that departed in November or December:

```
filter(flights, month == 11 | month == 12)
```

The order of operations doesn't work like English. You can't write `filter(flights, month == 11 | 12)`, which you might literally translate into "finds all flights that departed in November or December." Instead it finds all months that equal `11 | 12`, an expression that evaluates to `TRUE`. In a numeric context (like here), `TRUE` becomes one, so this finds all flights in January, not November or December. This is quite confusing!

A useful shorthand for this problem is `x %in% y`. This will select every row where `x` is one of the values in `y`. We could use it to rewrite the preceding code:

```
nov_dec <- filter(flights, month %in% c(11, 12))
```

Sometimes you can simplify complicated subsetting by remembering De Morgan's law: `!(x & y)` is the same as `!x | !y`, and `!(x | y)` is the same as `!x & !y`. For example, if you wanted to find flights that weren't delayed (on arrival or departure) by more than two hours, you could use either of the following two filters:

```
filter(flights, !(arr_delay > 120 | dep_delay > 120))
filter(flights, arr_delay <= 120, dep_delay <= 120)
```

As well as `&` and `|`, R also has `&&` and `||`. Don't use them here! You'll learn when you should use them in "Conditional Execution" on page 276.

Whenever you start using complicated, multipart expressions in `filter()`, consider making them explicit variables instead. That makes it much easier to check your work. You'll learn how to create new variables shortly.

Missing Values

One important feature of R that can make comparison tricky is missing values, or `NA`s ("not availables"). `NA` represents an unknown value so missing values are "contagious"; almost any operation involving an unknown value will also be unknown:

```
NA > 5
#> [1] NA
10 == NA
#> [1] NA
NA + 10
#> [1] NA
```

```
NA / 2
#> [1] NA
```

The most confusing result is this one:

```
NA == NA
#> [1] NA
```

It's easiest to understand why this is true with a bit more context:

```
# Let x be Mary's age. We don't know how old she is.
x <- NA

# Let y be John's age. We don't know how old he is.
y <- NA

# Are John and Mary the same age?
x == y
#> [1] NA
# We don't know!
```

If you want to determine if a value is missing, use `is.na()`:

```
is.na(x)
#> [1] TRUE
```

`filter()` only includes rows where the condition is `TRUE`; it excludes both `FALSE` and `NA` values. If you want to preserve missing values, ask for them explicitly:

```
df <- tibble(x = c(1, NA, 3))
filter(df, x > 1)
#> # A tibble: 1 × 1
#>       x
#>   <dbl>
#> 1     3
filter(df, is.na(x) | x > 1)
#> # A tibble: 2 × 1
#>       x
#>   <dbl>
#> 1    NA
#> 2     3
```

Exercises

1. Find all flights that:
   a. Had an arrival delay of two or more hours
   b. Flew to Houston (`IAH` or `HOU`)
   c. Were operated by United, American, or Delta

d. Departed in summer (July, August, and September)

e. Arrived more than two hours late, but didn't leave late

f. Were delayed by at least an hour, but made up over 30 minutes in flight

g. Departed between midnight and 6 a.m. (inclusive)

2. Another useful **dplyr** filtering helper is `between()`. What does it do? Can you use it to simplify the code needed to answer the previous challenges?

3. How many flights have a missing `dep_time`? What other variables are missing? What might these rows represent?

4. Why is `NA ^ 0` not missing? Why is `NA | TRUE` not missing? Why is `FALSE & NA` not missing? Can you figure out the general rule? (`NA * 0` is a tricky counterexample!)

Arrange Rows with arrange()

`arrange()` works similarly to `filter()` except that instead of selecting rows, it changes their order. It takes a data frame and a set of column names (or more complicated expressions) to order by. If you provide more than one column name, each additional column will be used to break ties in the values of preceding columns:

```
arrange(flights, year, month, day)
#> # A tibble: 336,776 × 19
#>    year month   day dep_time sched_dep_time dep_delay
#>   <int> <int> <int>    <int>          <int>     <dbl>
#> 1  2013     1     1      517            515         2
#> 2  2013     1     1      533            529         4
#> 3  2013     1     1      542            540         2
#> 4  2013     1     1      544            545        -1
#> 5  2013     1     1      554            600        -6
#> 6  2013     1     1      554            558        -4
#> # ... with 3.368e+05 more rows, and 13 more variables:
#> #   arr_time <int>, sched_arr_time <int>, arr_delay <dbl>,
#> #   carrier <chr>, flight <int>, tailnum <chr>, origin <chr>,
#> #   dest <chr>, air_time <dbl>, distance <dbl>, hour <dbl>,
#> #   minute <dbl>, time_hour <dttm>
```

Use `desc()` to reorder by a column in descending order:

```
arrange(flights, desc(arr_delay))
#> # A tibble: 336,776 × 19
#>    year month   day dep_time sched_dep_time dep_delay
```

```
#>   <int> <int> <int>    <int>          <int>     <dbl>
#> 1  2013     1     9      641            900      1301
#> 2  2013     6    15     1432           1935      1137
#> 3  2013     1    10     1121           1635      1126
#> 4  2013     9    20     1139           1845      1014
#> 5  2013     7    22      845           1600      1005
#> 6  2013     4    10     1100           1900       960
#> # ... with 3.368e+05 more rows, and 13 more variables:
#> #   arr_time <int>, sched_arr_time <int>, arr_delay <dbl>,
#> #   carrier <chr>, flight <int>, tailnum <chr>, origin <chr>,
#> #   dest <chr>, air_time <dbl>, distance <dbl>, hour <dbl>,
#> #   minute <dbl>, time_hour <dttm>,
```

Missing values are always sorted at the end:

```
df <- tibble(x = c(5, 2, NA))
arrange(df, x)
#> # A tibble: 3 × 1
#>       x
#>   <dbl>
#> 1     2
#> 2     5
#> 3    NA
arrange(df, desc(x))
#> # A tibble: 3 × 1
#>       x
#>   <dbl>
#> 1     5
#> 2     2
#> 3    NA
```

Exercises

1. How could you use `arrange()` to sort all missing values to the start? (Hint: use `is.na()`.)
2. Sort `flights` to find the most delayed flights. Find the flights that left earliest.
3. Sort `flights` to find the fastest flights.
4. Which flights traveled the longest? Which traveled the shortest?

Select Columns with select()

It's not uncommon to get datasets with hundreds or even thousands of variables. In this case, the first challenge is often narrowing in on the variables you're actually interested in. `select()` allows you to

rapidly zoom in on a useful subset using operations based on the names of the variables.

`select()` is not terribly useful with the flight data because we only have 19 variables, but you can still get the general idea:

```
# Select columns by name
select(flights, year, month, day)
#> # A tibble: 336,776 × 3
#>    year month   day
#>   <int> <int> <int>
#> 1  2013     1     1
#> 2  2013     1     1
#> 3  2013     1     1
#> 4  2013     1     1
#> 5  2013     1     1
#> 6  2013     1     1
#> # ... with 3.368e+05 more rows

# Select all columns between year and day (inclusive)
select(flights, year:day)
#> # A tibble: 336,776 × 3
#>    year month   day
#>   <int> <int> <int>
#> 1  2013     1     1
#> 2  2013     1     1
#> 3  2013     1     1
#> 4  2013     1     1
#> 5  2013     1     1
#> 6  2013     1     1
#> # ... with 3.368e+05 more rows

# Select all columns except those from year to day (inclusive)
select(flights, -(year:day))
#> # A tibble: 336,776 × 16
#>   dep_time sched_dep_time dep_delay arr_time sched_arr_time
#>      <int>          <int>     <dbl>    <int>          <int>
#> 1      517            515         2      830            819
#> 2      533            529         4      850            830
#> 3      542            540         2      923            850
#> 4      544            545        -1     1004           1022
#> 5      554            600        -6      812            837
#> 6      554            558        -4      740            728
#> # ... with 3.368e+05 more rows, and 12 more variables:
#> #   arr_delay <dbl>, carrier <chr>, flight <int>,
#> #   tailnum <chr>, origin <chr>, dest <chr>, air_time <dbl>,
#> #   distance <dbl>,  hour <dbl>, minute <dbl>,
#> #   time_hour <dttm>
```

There are a number of helper functions you can use within `select()`:

- `starts_with("abc")` matches names that begin with "abc".
- `ends_with("xyz")` matches names that end with "xyz".
- `contains("ijk")` matches names that contain "ijk".
- `matches("(.)\\1")` selects variables that match a regular expression. This one matches any variables that contain repeated characters. You'll learn more about regular expressions in Chapter 11.
- `num_range("x", 1:3)` matches `x1`, `x2`, and `x3`.

See `?select` for more details.

`select()` can be used to rename variables, but it's rarely useful because it drops all of the variables not explicitly mentioned. Instead, use `rename()`, which is a variant of `select()` that keeps all the variables that aren't explicitly mentioned:

```
rename(flights, tail_num = tailnum)
#> # A tibble: 336,776 × 19
#>    year month   day dep_time sched_dep_time dep_delay
#>   <int> <int> <int>    <int>          <int>     <dbl>
#> 1  2013     1     1      517            515         2
#> 2  2013     1     1      533            529         4
#> 3  2013     1     1      542            540         2
#> 4  2013     1     1      544            545        -1
#> 5  2013     1     1      554            600        -6
#> 6  2013     1     1      554            558        -4
#> # ... with 3.368e+05 more rows, and 13 more variables:
#> #   arr_time <int>, sched_arr_time <int>, arr_delay <dbl>,
#> #   carrier <chr>, flight <int>, tail_num <chr>,
#> #   origin <chr>, dest <chr>, air_time <dbl>,
#> #   distance <dbl>, hour <dbl>, minute <dbl>,
#> #   time_hour <dttm>
```

Another option is to use `select()` in conjunction with the `everything()` helper. This is useful if you have a handful of variables you'd like to move to the start of the data frame:

```
select(flights, time_hour, air_time, everything())
#> # A tibble: 336,776 × 19
#>             time_hour air_time  year month   day dep_time
#>                <dttm>    <dbl> <int> <int> <int>    <int>
#> 1 2013-01-01 05:00:00      227  2013     1     1      517
#> 2 2013-01-01 05:00:00      227  2013     1     1      533
#> 3 2013-01-01 05:00:00      160  2013     1     1      542
#> 4 2013-01-01 05:00:00      183  2013     1     1      544
#> 5 2013-01-01 06:00:00      116  2013     1     1      554
```

```
#> 6 2013-01-01 05:00:00      150  2013     1     1      554
#> # ... with 3.368e+05 more rows, and 13 more variables:
#> #   sched_dep_time <int>, dep_delay <dbl>, arr_time <int>,
#> #   sched_arr_time <int>, arr_delay <dbl>, carrier <chr>,
#> #   flight <int>, tailnum <chr>, origin <chr>, dest <chr>,
#> #   distance <dbl>, hour <dbl>, minute <dbl>
```

Exercises

1. Brainstorm as many ways as possible to select `dep_time`, `dep_delay`, `arr_time`, and `arr_delay` from `flights`.
2. What happens if you include the name of a variable multiple times in a `select()` call?
3. What does the `one_of()` function do? Why might it be helpful in conjunction with this vector?

```
vars <- c(
  "year", "month", "day", "dep_delay", "arr_delay"
)
```

4. Does the result of running the following code surprise you? How do the select helpers deal with case by default? How can you change that default?

```
select(flights, contains("TIME"))
```

Add New Variables with mutate()

Besides selecting sets of existing columns, it's often useful to add new columns that are functions of existing columns. That's the job of `mutate()`.

`mutate()` always adds new columns at the end of your dataset so we'll start by creating a narrower dataset so we can see the new variables. Remember that when you're in RStudio, the easiest way to see all the columns is `View()`:

```
flights_sml <- select(flights,
  year:day,
  ends_with("delay"),
  distance,
  air_time
)
mutate(flights_sml,
  gain = arr_delay - dep_delay,
  speed = distance / air_time * 60
```

```
)
#> # A tibble: 336,776 × 9
#>    year month   day dep_delay arr_delay distance air_time
#>   <int> <int> <int>     <dbl>     <dbl>    <dbl>    <dbl>
#> 1  2013     1     1         2        11     1400      227
#> 2  2013     1     1         4        20     1416      227
#> 3  2013     1     1         2        33     1089      160
#> 4  2013     1     1        -1       -18     1576      183
#> 5  2013     1     1        -6       -25      762      116
#> 6  2013     1     1        -4        12      719      150
#> # ... with 3.368e+05 more rows, and 2 more variables:
#> #   gain <dbl>, speed <dbl>
```

Note that you can refer to columns that you've just created:

```
mutate(flights_sml,
  gain = arr_delay - dep_delay,
  hours = air_time / 60,
  gain_per_hour = gain / hours
)
#> # A tibble: 336,776 × 10
#>    year month   day dep_delay arr_delay distance air_time
#>   <int> <int> <int>     <dbl>     <dbl>    <dbl>    <dbl>
#> 1  2013     1     1         2        11     1400      227
#> 2  2013     1     1         4        20     1416      227
#> 3  2013     1     1         2        33     1089      160
#> 4  2013     1     1        -1       -18     1576      183
#> 5  2013     1     1        -6       -25      762      116
#> 6  2013     1     1        -4        12      719      150
#> # ... with 3.368e+05 more rows, and 3 more variables:
#> #   gain <dbl>, hours <dbl>, gain_per_hour <dbl>
```

If you only want to keep the new variables, use `transmute()`:

```
transmute(flights,
  gain = arr_delay - dep_delay,
  hours = air_time / 60,
  gain_per_hour = gain / hours
)
#> # A tibble: 336,776 × 3
#>    gain hours gain_per_hour
#>   <dbl> <dbl>         <dbl>
#> 1     9  3.78          2.38
#> 2    16  3.78          4.23
#> 3    31  2.67         11.62
#> 4   -17  3.05         -5.57
#> 5   -19  1.93         -9.83
#> 6    16  2.50          6.40
#> # ... with 3.368e+05 more rows
```

Useful Creation Functions

There are many functions for creating new variables that you can use with `mutate()`. The key property is that the function must be vectorized: it must take a vector of values as input, and return a vector with the same number of values as output. There's no way to list every possible function that you might use, but here's a selection of functions that are frequently useful:

Arithmetic operators `+`, `-`, `*`, `/`, `^`
: These are all vectorized, using the so-called "recycling rules." If one parameter is shorter than the other, it will be automatically extended to be the same length. This is most useful when one of the arguments is a single number: `air_time / 60`, `hours * 60 + minute`, etc.

  Arithmetic operators are also useful in conjunction with the aggregate functions you'll learn about later. For example, `x / sum(x)` calculates the proportion of a total, and `y - mean(y)` computes the difference from the mean.

Modular arithmetic (`%/%` and `%%`)
: `%/%` (integer division) and `%%` (remainder), where `x == y * (x %/% y) + (x %% y)`. Modular arithmetic is a handy tool because it allows you to break integers into pieces. For example, in the flights dataset, you can compute `hour` and `minute` from `dep_time` with:

```
transmute(flights,
  dep_time,
  hour = dep_time %/% 100,
  minute = dep_time %% 100
)
#> # A tibble: 336,776 × 3
#>   dep_time  hour minute
#>      <int> <dbl>  <dbl>
#> 1      517     5     17
#> 2      533     5     33
#> 3      542     5     42
#> 4      544     5     44
#> 5      554     5     54
#> 6      554     5     54
#> # ... with 3.368e+05 more rows
```

Logs `log()`, `log2()`, `log10()`
: Logarithms are an incredibly useful transformation for dealing with data that ranges across multiple orders of magnitude. They also convert multiplicative relationships to additive, a feature we'll come back to in Part IV.

  All else being equal, I recommend using `log2()` because it's easy to interpret: a difference of 1 on the log scale corresponds to doubling on the original scale and a difference of –1 corresponds to halving.

Offsets
: `lead()` and `lag()` allow you to refer to leading or lagging values. This allows you to compute running differences (e.g., `x - lag(x)`) or find when values change (`x != lag(x)`). They are most useful in conjunction with `group_by()`, which you'll learn about shortly:

```
(x <- 1:10)
#>  [1]  1  2  3  4  5  6  7  8  9 10
lag(x)
#>  [1] NA  1  2  3  4  5  6  7  8  9
lead(x)
#>  [1]  2  3  4  5  6  7  8  9 10 NA
```

Cumulative and rolling aggregates
: R provides functions for running sums, products, mins, and maxes: `cumsum()`, `cumprod()`, `cummin()`, `cummax()`; and **dplyr** provides `cummean()` for cumulative means. If you need rolling aggregates (i.e., a sum computed over a rolling window), try the **RcppRoll** package:

```
x
#>  [1]  1  2  3  4  5  6  7  8  9 10
cumsum(x)
#>  [1]  1  3  6 10 15 21 28 36 45 55
cummean(x)
#>  [1] 1.0 1.5 2.0 2.5 3.0 3.5 4.0 4.5 5.0 5.5
```

Logical comparisons `<`, `<=`, `>`, `>=`, `!=`
: If you're doing a complex sequence of logical operations it's often a good idea to store the interim values in new variables so you can check that each step is working as expected.

Ranking

There are a number of ranking functions, but you should start with `min_rank()`. It does the most usual type of ranking (e.g., first, second, third, fourth). The default gives the smallest values the smallest ranks; use `desc(x)` to give the largest values the smallest ranks:

```
y <- c(1, 2, 2, NA, 3, 4)
min_rank(y)
#> [1]  1  2  2 NA  4  5
min_rank(desc(y))
#> [1]  5  3  3 NA  2  1
```

If `min_rank()` doesn't do what you need, look at the variants `row_number()`, `dense_rank()`, `percent_rank()`, `cume_dist()`, and `ntile()`. See their help pages for more details:

```
row_number(y)
#> [1]  1  2  3 NA  4  5
dense_rank(y)
#> [1]  1  2  2 NA  3  4
percent_rank(y)
#> [1] 0.00 0.25 0.25   NA 0.75 1.00
cume_dist(y)
#> [1] 0.2 0.6 0.6  NA 0.8 1.0
```

Exercises

1. Currently `dep_time` and `sched_dep_time` are convenient to look at, but hard to compute with because they're not really continuous numbers. Convert them to a more convenient representation of number of minutes since midnight.
2. Compare `air_time` with `arr_time - dep_time`. What do you expect to see? What do you see? What do you need to do to fix it?
3. Compare `dep_time`, `sched_dep_time`, and `dep_delay`. How would you expect those three numbers to be related?
4. Find the 10 most delayed flights using a ranking function. How do you want to handle ties? Carefully read the documentation for `min_rank()`.
5. What does `1:3 + 1:10` return? Why?
6. What trigonometric functions does R provide?

Grouped Summaries with summarize()

The last key verb is `summarize()`. It collapses a data frame to a single row:

```
summarize(flights, delay = mean(dep_delay, na.rm = TRUE))
#> # A tibble: 1 × 1
#>   delay
#>   <dbl>
#> 1  12.6
```

(We'll come back to what that `na.rm = TRUE` means very shortly.)

`summarize()` is not terribly useful unless we pair it with `group_by()`. This changes the unit of analysis from the complete dataset to individual groups. Then, when you use the **dplyr** verbs on a grouped data frame they'll be automatically applied "by group." For example, if we applied exactly the same code to a data frame grouped by date, we get the average delay per date:

```
by_day <- group_by(flights, year, month, day)
summarize(by_day, delay = mean(dep_delay, na.rm = TRUE))
#> Source: local data frame [365 x 4]
#> Groups: year, month [?]
#>
#>    year month   day delay
#>   <int> <int> <int> <dbl>
#> 1  2013     1     1 11.55
#> 2  2013     1     2 13.86
#> 3  2013     1     3 10.99
#> 4  2013     1     4  8.95
#> 5  2013     1     5  5.73
#> 6  2013     1     6  7.15
#> # ... with 359 more rows
```

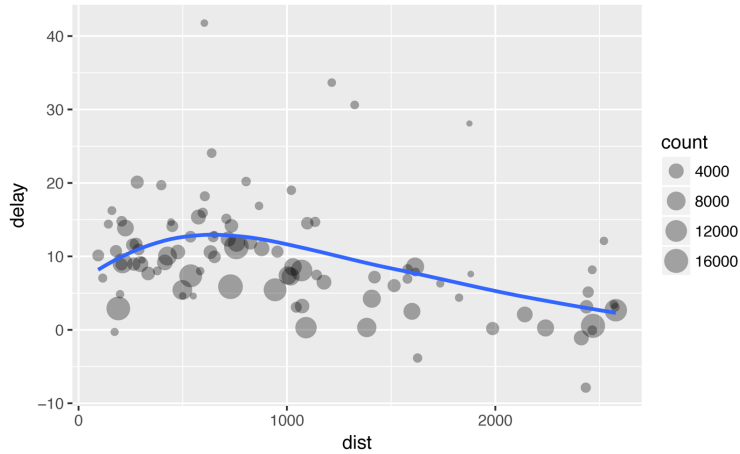
Together `group_by()` and `summarize()` provide one of the tools that you'll use most commonly when working with **dplyr**: grouped summaries. But before we go any further with this, we need to introduce a powerful new idea: the pipe.

Combining Multiple Operations with the Pipe

Imagine that we want to explore the relationship between the distance and average delay for each location. Using what you know about **dplyr**, you might write code like this:

```
by_dest <- group_by(flights, dest)
delay <- summarize(by_dest,
  count = n(),
```

```
  dist = mean(distance, na.rm = TRUE),
  delay = mean(arr_delay, na.rm = TRUE)
)
delay <- filter(delay, count > 20, dest != "HNL")

# It looks like delays increase with distance up to ~750 miles
# and then decrease. Maybe as flights get longer there's more
# ability to make up delays in the air?
ggplot(data = delay, mapping = aes(x = dist, y = delay)) +
  geom_point(aes(size = count), alpha = 1/3) +
  geom_smooth(se = FALSE)
#> `geom_smooth()` using method = 'loess'
```



There are three steps to prepare this data:

1. Group flights by destination.
2. Summarize to compute distance, average delay, and number of flights.
3. Filter to remove noisy points and Honolulu airport, which is almost twice as far away as the next closest airport.

This code is a little frustrating to write because we have to give each intermediate data frame a name, even though we don't care about it. Naming things is hard, so this slows down our analysis.

There's another way to tackle the same problem with the pipe, `%>%`:

```
delays <- flights %>%
  group_by(dest) %>%
  summarize(
    count = n(),
```

```
    dist = mean(distance, na.rm = TRUE),
    delay = mean(arr_delay, na.rm = TRUE)
  ) %>%
  filter(count > 20, dest != "HNL")
```

This focuses on the transformations, not what's being transformed, which makes the code easier to read. You can read it as a series of imperative statements: group, then summarize, then filter. As suggested by this reading, a good way to pronounce `%>%` when reading code is "then."

Behind the scenes, `x %>% f(y)` turns into `f(x, y)`, and `x %>% f(y) %>% g(z)` turns into `g(f(x, y), z)`, and so on. You can use the pipe to rewrite multiple operations in a way that you can read left-to-right, top-to-bottom. We'll use piping frequently from now on because it considerably improves the readability of code, and we'll come back to it in more detail in Chapter 14.

Working with the pipe is one of the key criteria for belonging to the tidyverse. The only exception is **ggplot2**: it was written before the pipe was discovered. Unfortunately, the next iteration of **ggplot2**, **ggvis**, which does use the pipe, isn't ready for prime time yet.

Missing Values

You may have wondered about the `na.rm` argument we used earlier. What happens if we don't set it?

```
flights %>%
  group_by(year, month, day) %>%
  summarize(mean = mean(dep_delay))
#> Source: local data frame [365 x 4]
#> Groups: year, month [?]
#>
#>    year month   day  mean
#>   <int> <int> <int> <dbl>
#> 1  2013     1     1    NA
#> 2  2013     1     2    NA
#> 3  2013     1     3    NA
#> 4  2013     1     4    NA
#> 5  2013     1     5    NA
#> 6  2013     1     6    NA
#> # ... with 359 more rows
```

We get a lot of missing values! That's because aggregation functions obey the usual rule of missing values: if there's any missing value in the input, the output will be a missing value. Fortunately, all aggre-

gation functions have an `na.rm` argument, which removes the missing values prior to computation:

```
flights %>%
  group_by(year, month, day) %>%
  summarize(mean = mean(dep_delay, na.rm = TRUE))
#> Source: local data frame [365 x 4]
#> Groups: year, month [?]
#>
#>    year month   day  mean
#>   <int> <int> <int> <dbl>
#> 1  2013     1     1 11.55
#> 2  2013     1     2 13.86
#> 3  2013     1     3 10.99
#> 4  2013     1     4  8.95
#> 5  2013     1     5  5.73
#> 6  2013     1     6  7.15
#> # ... with 359 more rows
```

In this case, where missing values represent cancelled flights, we could also tackle the problem by first removing the cancelled flights. We'll save this dataset so we can reuse it in the next few examples:

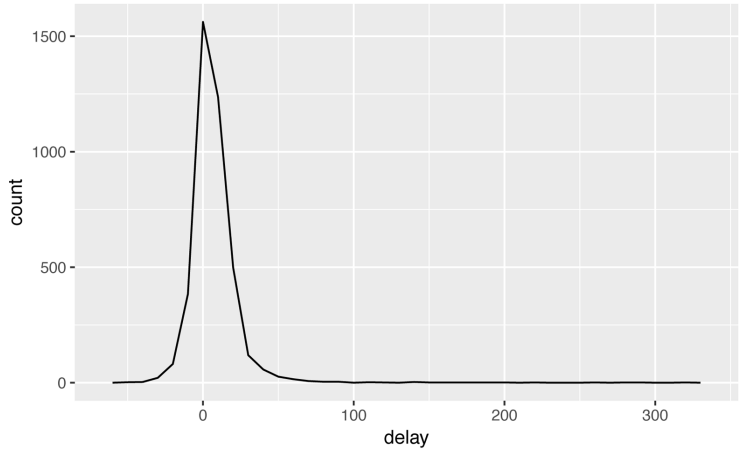
```
not_cancelled <- flights %>%
  filter(!is.na(dep_delay), !is.na(arr_delay))

not_cancelled %>%
  group_by(year, month, day) %>%
  summarize(mean = mean(dep_delay))
#> Source: local data frame [365 x 4]
#> Groups: year, month [?]
#>
#>    year month   day  mean
#>   <int> <int> <int> <dbl>
#> 1  2013     1     1 11.44
#> 2  2013     1     2 13.68
#> 3  2013     1     3 10.91
#> 4  2013     1     4  8.97
#> 5  2013     1     5  5.73
#> 6  2013     1     6  7.15
#> # ... with 359 more rows
```

Counts

Whenever you do any aggregation, it's always a good idea to include either a count (`n()`), or a count of nonmissing values (`sum(!is.na(x))`). That way you can check that you're not drawing conclusions based on very small amounts of data. For example, let's

look at the planes (identified by their tail number) that have the highest average delays:

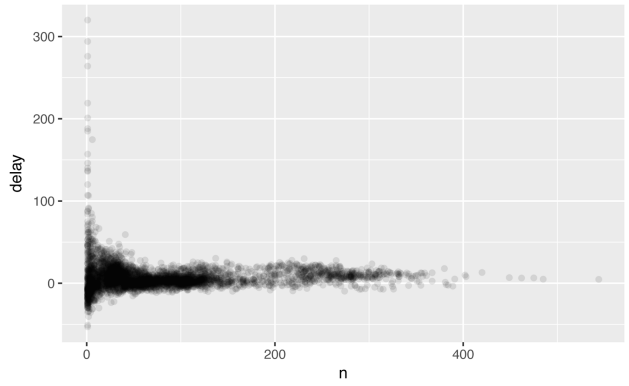
```
delays <- not_cancelled %>%
  group_by(tailnum) %>%
  summarize(
    delay = mean(arr_delay)
  )

ggplot(data = delays, mapping = aes(x = delay)) +
  geom_freqpoly(binwidth = 10)
```



Wow, there are some planes that have an *average* delay of 5 hours (300 minutes)!

The story is actually a little more nuanced. We can get more insight if we draw a scatterplot of number of flights versus average delay:

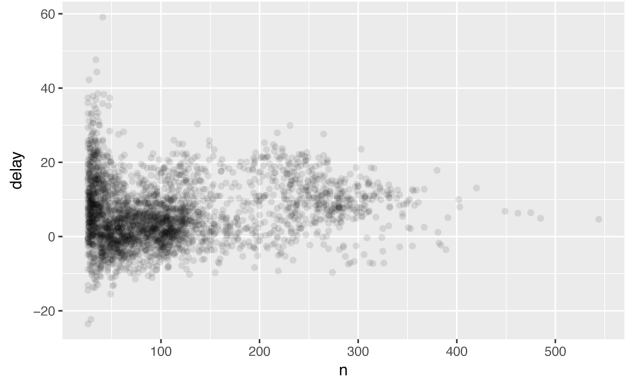
```
delays <- not_cancelled %>%
  group_by(tailnum) %>%
  summarize(
    delay = mean(arr_delay, na.rm = TRUE),
    n = n()
  )

ggplot(data = delays, mapping = aes(x = n, y = delay)) +
  geom_point(alpha = 1/10)
```



Not surprisingly, there is much greater variation in the average delay when there are few flights. The shape of this plot is very characteristic: whenever you plot a mean (or other summary) versus group size, you'll see that the variation decreases as the sample size increases.

When looking at this sort of plot, it's often useful to filter out the groups with the smallest numbers of observations, so you can see more of the pattern and less of the extreme variation in the smallest groups. This is what the following code does, as well as showing you a handy pattern for integrating **ggplot2** into **dplyr** flows. It's a bit painful that you have to switch from `%>%` to `+`, but once you get the hang of it, it's quite convenient:

```
delays %>%
  filter(n > 25) %>%
  ggplot(mapping = aes(x = n, y = delay)) +
    geom_point(alpha = 1/10)
```



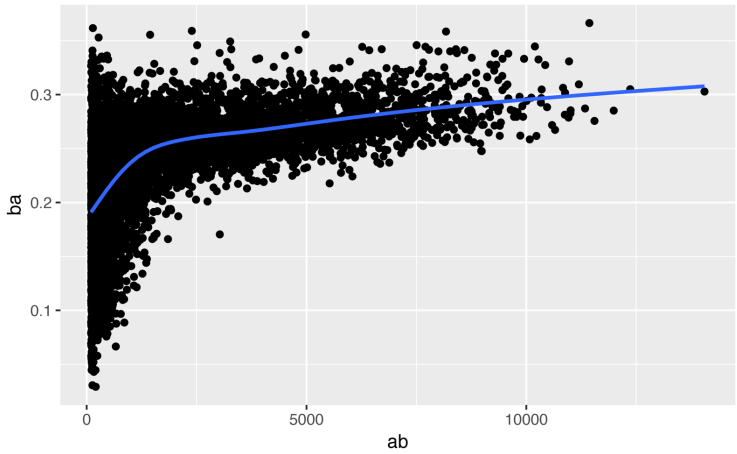
RStudio tip: a useful keyboard shortcut is Cmd/Ctrl-Shift-P. This resends the previously sent chunk from the editor to the console. This is very convenient when you're (e.g.) exploring the value of `n` in the preceding example. You send the whole block once with Cmd/Ctrl-Enter, then you modify the value of `n` and press Cmd/Ctrl-Shift-P to resend the complete block.

There's another common variation of this type of pattern. Let's look at how the average performance of batters in baseball is related to the number of times they're at bat. Here I use data from the **Lahman** package to compute the batting average (number of hits / number of attempts) of every major league baseball player.

When I plot the skill of the batter (measured by the batting average, `ba`) against the number of opportunities to hit the ball (measured by at bat, `ab`), you see two patterns:

- As above, the variation in our aggregate decreases as we get more data points.
- There's a positive correlation between skill (`ba`) and opportunities to hit the ball (`ab`). This is because teams control who gets to play, and obviously they'll pick their best players:

```
# Convert to a tibble so it prints nicely
batting <- as_tibble(Lahman::Batting)

batters <- batting %>%
  group_by(playerID) %>%
  summarize(
    ba = sum(H, na.rm = TRUE) / sum(AB, na.rm = TRUE),
    ab = sum(AB, na.rm = TRUE)
  )

batters %>%
  filter(ab > 100) %>%
  ggplot(mapping = aes(x = ab, y = ba)) +
    geom_point() +
    geom_smooth(se = FALSE)
#> `geom_smooth()` using method = 'gam'
```



This also has important implications for ranking. If you naively sort on `desc(ba)`, the people with the best batting averages are clearly lucky, not skilled:

```
batters %>%
  arrange(desc(ba))
#> # A tibble: 18,659 × 3
#>    playerID    ba    ab
#>       <chr> <dbl> <int>
#> 1 abramge01     1     1
#> 2 banisje01     1     1
#> 3 bartocl01     1     1
#> 4  bassdo01     1     1
#> 5 birasst01     1     2
#> 6 bruneju01     1     1
#> # ... with 1.865e+04 more rows
```

You can find a good explanation of this problem at *http://bit.ly/Bayesbbal* and *http://bit.ly/notsortavg*.

Useful Summary Functions

Just using means, counts, and sum can get you a long way, but R provides many other useful summary functions:

Measures of location

We've used `mean(x)`, but `median(x)` is also useful. The mean is the sum divided by the length; the median is a value where 50% of x is above it, and 50% is below it.

It's sometimes useful to combine aggregation with logical subsetting. We haven't talked about this sort of subsetting yet, but you'll learn more about it in "Subsetting" on page 304:

```
not_cancelled %>%
  group_by(year, month, day) %>%
  summarize(
    # average delay:
    avg_delay1 = mean(arr_delay),
    # average positive delay:
    avg_delay2 = mean(arr_delay[arr_delay > 0])
  )
#> Source: local data frame [365 x 5]
#> Groups: year, month [?]
#>
#>    year month   day avg_delay1 avg_delay2
#>   <int> <int> <int>      <dbl>      <dbl>
#> 1  2013     1     1      12.65       32.5
#> 2  2013     1     2      12.69       32.0
#> 3  2013     1     3       5.73       27.7
#> 4  2013     1     4      -1.93       28.3
#> 5  2013     1     5      -1.53       22.6
#> 6  2013     1     6       4.24       24.4
#> # ... with 359 more rows
```

Measures of spread `sd(x)`, `IQR(x)`, `mad(x)`

The mean squared deviation, or standard deviation or sd for short, is the standard measure of spread. The interquartile range `IQR()` and median absolute deviation `mad(x)` are robust equivalents that may be more useful if you have outliers:

```
# Why is distance to some destinations more variable
# than to others?
not_cancelled %>%
  group_by(dest) %>%
  summarize(distance_sd = sd(distance)) %>%
  arrange(desc(distance_sd))
#> # A tibble: 104 × 2
#>    dest distance_sd
#>   <chr>       <dbl>
#> 1   EGE       10.54
#> 2   SAN       10.35
#> 3   SFO       10.22
#> 4   HNL       10.00
#> 5   SEA        9.98
#> 6   LAS        9.91
#> # ... with 98 more rows
```

Measures of rank `min(x)`, `quantile(x, 0.25)`, `max(x)`

Quantiles are a generalization of the median. For example, `quantile(x, 0.25)` will find a value of `x` that is greater than 25% of the values, and less than the remaining 75%:

```
# When do the first and last flights leave each day?
not_cancelled %>%
  group_by(year, month, day) %>%
  summarize(
    first = min(dep_time),
    last = max(dep_time)
  )
#> Source: local data frame [365 x 5]
#> Groups: year, month [?]
#>
#>    year month   day first  last
#>   <int> <int> <int> <int> <int>
#> 1  2013     1     1   517  2356
#> 2  2013     1     2    42  2354
#> 3  2013     1     3    32  2349
#> 4  2013     1     4    25  2358
#> 5  2013     1     5    14  2357
#> 6  2013     1     6    16  2355
#> # ... with 359 more rows
```

Measures of position `first(x)`, `nth(x, 2)`, `last(x)`

These work similarly to `x[1]`, `x[2]`, and `x[length(x)]` but let you set a default value if that position does not exist (i.e., you're trying to get the third element from a group that only has two elements). For example, we can find the first and last departure for each day:

```
not_cancelled %>%
  group_by(year, month, day) %>%
  summarize(
    first_dep = first(dep_time),
    last_dep = last(dep_time)
  )
#> Source: local data frame [365 x 5]
#> Groups: year, month [?]
#>
#>    year month   day first_dep last_dep
#>   <int> <int> <int>     <int>    <int>
#> 1  2013     1     1       517     2356
#> 2  2013     1     2        42     2354
#> 3  2013     1     3        32     2349
#> 4  2013     1     4        25     2358
#> 5  2013     1     5        14     2357
```

```
#> 6  2013     1     6        16     2355
#> # ... with 359 more rows
```

These functions are complementary to filtering on ranks. Filtering gives you all variables, with each observation in a separate row:

```
not_cancelled %>%
  group_by(year, month, day) %>%
  mutate(r = min_rank(desc(dep_time))) %>%
  filter(r %in% range(r))
#> Source: local data frame [770 x 20]
#> Groups: year, month, day [365]
#>
#>    year month   day dep_time sched_dep_time dep_delay
#>   <int> <int> <int>    <int>          <int>     <dbl>
#> 1  2013     1     1      517            515         2
#> 2  2013     1     1     2356           2359        -3
#> 3  2013     1     2       42           2359        43
#> 4  2013     1     2     2354           2359        -5
#> 5  2013     1     3       32           2359        33
#> 6  2013     1     3     2349           2359       -10
#> # ... with 764 more rows, and 13 more variables:
#> #   arr_time <int>, sched_arr_time <int>,
#> #   arr_delay <dbl>, carrier <chr>, flight <int>,
#> #   tailnum <chr>, origin <chr>, dest <chr>,
#> #   air_time <dbl>, distance <dbl>, hour <dbl>,
#> #   minute <dbl>, time_hour <dttm>, r <int>
```

Counts

You've seen `n()`, which takes no arguments, and returns the size of the current group. To count the number of non-missing values, use `sum(!is.na(x))`. To count the number of distinct (unique) values, use `n_distinct(x)`:

```
# Which destinations have the most carriers?
not_cancelled %>%
  group_by(dest) %>%
  summarize(carriers = n_distinct(carrier)) %>%
  arrange(desc(carriers))
#> # A tibble: 104 × 2
#>    dest carriers
#>   <chr>    <int>
#> 1   ATL        7
#> 2   BOS        7
#> 3   CLT        7
#> 4   ORD        7
#> 5   TPA        7
```

```
#> 6   AUS        6
#> # ... with 98 more rows
```

Counts are so useful that **dplyr** provides a simple helper if all you want is a count:

```
not_cancelled %>%
  count(dest)
#> # A tibble: 104 × 2
#>    dest     n
#>   <chr> <int>
#> 1   ABQ   254
#> 2   ACK   264
#> 3   ALB   418
#> 4   ANC     8
#> 5   ATL 16837
#> 6   AUS  2411
#> # ... with 98 more rows
```

You can optionally provide a weight variable. For example, you could use this to "count" (sum) the total number of miles a plane flew:

```
not_cancelled %>%
  count(tailnum, wt = distance)
#> # A tibble: 4,037 × 2
#>   tailnum      n
#>     <chr>  <dbl>
#> 1  D942DN   3418
#> 2  N0EGMQ 239143
#> 3  N10156 109664
#> 4  N102UW  25722
#> 5  N103US  24619
#> 6  N104UW  24616
#> # ... with 4,031 more rows
```

Counts and proportions of logical values `sum(x > 10)`, `mean(y == 0)`

When used with numeric functions, `TRUE` is converted to 1 and `FALSE` to 0. This makes `sum()` and `mean()` very useful: `sum(x)` gives the number of `TRUE`s in `x`, and `mean(x)` gives the proportion:

```
# How many flights left before 5am? (these usually
# indicate delayed flights from the previous day)
not_cancelled %>%
  group_by(year, month, day) %>%
  summarize(n_early = sum(dep_time < 500))
#> Source: local data frame [365 x 4]
#> Groups: year, month [?]
```

```
#>
#>    year month   day n_early
#>   <int> <int> <int>   <int>
#> 1  2013     1     1       0
#> 2  2013     1     2       3
#> 3  2013     1     3       4
#> 4  2013     1     4       3
#> 5  2013     1     5       3
#> 6  2013     1     6       2
#> # ... with 359 more rows

# What proportion of flights are delayed by more
# than an hour?
not_cancelled %>%
  group_by(year, month, day) %>%
  summarize(hour_perc = mean(arr_delay > 60))
#> Source: local data frame [365 x 4]
#> Groups: year, month [?]
#>
#>    year month   day hour_perc
#>   <int> <int> <int>     <dbl>
#> 1  2013     1     1    0.0722
#> 2  2013     1     2    0.0851
#> 3  2013     1     3    0.0567
#> 4  2013     1     4    0.0396
#> 5  2013     1     5    0.0349
#> 6  2013     1     6    0.0470
#> # ... with 359 more rows
```

Grouping by Multiple Variables

When you group by multiple variables, each summary peels off one level of the grouping. That makes it easy to progressively roll up a dataset:

```
daily <- group_by(flights, year, month, day)
(per_day   <- summarize(daily, flights = n()))
#> Source: local data frame [365 x 4]
#> Groups: year, month [?]
#>
#>    year month   day flights
#>   <int> <int> <int>   <int>
#> 1  2013     1     1     842
#> 2  2013     1     2     943
#> 3  2013     1     3     914
#> 4  2013     1     4     915
#> 5  2013     1     5     720
#> 6  2013     1     6     832
#> # ... with 359 more rows
```

```
(per_month <- summarize(per_day, flights = sum(flights)))
#> Source: local data frame [12 x 3]
#> Groups: year [?]
#>
#>    year month flights
#>   <int> <int>   <int>
#> 1  2013     1   27004
#> 2  2013     2   24951
#> 3  2013     3   28834
#> 4  2013     4   28330
#> 5  2013     5   28796
#> 6  2013     6   28243
#> # ... with 6 more rows

(per_year  <- summarize(per_month, flights = sum(flights)))
#> # A tibble: 1 × 2
#>    year flights
#>   <int>   <int>
#> 1  2013  336776
```

Be careful when progressively rolling up summaries: it's OK for sums and counts, but you need to think about weighting means and variances, and it's not possible to do it exactly for rank-based statistics like the median. In other words, the sum of groupwise sums is the overall sum, but the median of groupwise medians is not the overall median.

Ungrouping

If you need to remove grouping, and return to operations on ungrouped data, use `ungroup()`:

```
daily %>%
  ungroup() %>%             # no longer grouped by date
  summarize(flights = n())  # all flights
#> # A tibble: 1 × 1
#>   flights
#>     <int>
#> 1  336776
```

Exercises

1. Brainstorm at least five different ways to assess the typical delay characteristics of a group of flights. Consider the following scenarios:

- A flight is 15 minutes early 50% of the time, and 15 minutes late 50% of the time.
- A flight is always 10 minutes late.
- A flight is 30 minutes early 50% of the time, and 30 minutes late 50% of the time.
- 99% of the time a flight is on time. 1% of the time it's 2 hours late.

Which is more important: arrival delay or departure delay?

2. Come up with another approach that will give you the same output as `not_cancelled %>% count(dest)` and `not_cancelled %>% count(tailnum, wt = distance)` (without using `count()`).
3. Our definition of cancelled flights (`is.na(dep_delay) | is.na(arr_delay)`) is slightly suboptimal. Why? Which is the most important column?
4. Look at the number of cancelled flights per day. Is there a pattern? Is the proportion of cancelled flights related to the average delay?
5. Which carrier has the worst delays? Challenge: can you disentangle the effects of bad airports versus bad carriers? Why/why not? (Hint: think about `flights %>% group_by(carrier, dest) %>% summarize(n())`.)
6. For each plane, count the number of flights before the first delay of greater than 1 hour.
7. What does the `sort` argument to `count()` do? When might you use it?

Grouped Mutates (and Filters)

Grouping is most useful in conjunction with `summarize()`, but you can also do convenient operations with `mutate()` and `filter()`:

- Find the worst members of each group:

```
flights_sml %>%
  group_by(year, month, day) %>%
  filter(rank(desc(arr_delay)) < 10)
```

```
#> Source: local data frame [3,306 x 7]
#> Groups: year, month, day [365]
#>
#>    year month   day dep_delay arr_delay distance
#>   <int> <int> <int>     <dbl>     <dbl>    <dbl>
#> 1  2013     1     1       853       851      184
#> 2  2013     1     1       290       338     1134
#> 3  2013     1     1       260       263      266
#> 4  2013     1     1       157       174      213
#> 5  2013     1     1       216       222      708
#> 6  2013     1     1       255       250      589
#> # ... with 3,300 more rows, and 1 more variables:
#> #   air_time <dbl>
```

- Find all groups bigger than a threshold:

```
popular_dests <- flights %>%
  group_by(dest) %>%
  filter(n() > 365)
popular_dests
#> Source: local data frame [332,577 x 19]
#> Groups: dest [77]
#>
#>    year month   day dep_time sched_dep_time dep_delay
#>   <int> <int> <int>    <int>          <int>     <dbl>
#> 1  2013     1     1      517            515         2
#> 2  2013     1     1      533            529         4
#> 3  2013     1     1      542            540         2
#> 4  2013     1     1      544            545        -1
#> 5  2013     1     1      554            600        -6
#> 6  2013     1     1      554            558        -4
#> # ... with 3.326e+05 more rows, and 13 more variables:
#> #   arr_time <int>, sched_arr_time <int>,
#> #   arr_delay <dbl>, carrier <chr>, flight <int>,
#> #   tailnum <chr>, origin <chr>, dest <chr>,
#> #   air_time <dbl>, distance <dbl>, hour <dbl>,
#> #   minute <dbl>, time_hour <dttm>
```

- Standardize to compute per group metrics:

```
popular_dests %>%
  filter(arr_delay > 0) %>%
  mutate(prop_delay = arr_delay / sum(arr_delay)) %>%
  select(year:day, dest, arr_delay, prop_delay)
#> Source: local data frame [131,106 x 6]
#> Groups: dest [77]
#>
#>    year month   day  dest arr_delay prop_delay
#>   <int> <int> <int> <chr>     <dbl>      <dbl>
#> 1  2013     1     1   IAH        11   1.11e-04
```

```
#> 2  2013     1     1   IAH        20   2.01e-04
#> 3  2013     1     1   MIA        33   2.35e-04
#> 4  2013     1     1   ORD        12   4.24e-05
#> 5  2013     1     1   FLL        19   9.38e-05
#> 6  2013     1     1   ORD         8   2.83e-05
#> # ... with 1.311e+05 more rows
```

A grouped filter is a grouped mutate followed by an ungrouped filter. I generally avoid them except for quick-and-dirty manipulations: otherwise it's hard to check that you've done the manipulation correctly.

Functions that work most naturally in grouped mutates and filters are known as window functions (versus the summary functions used for summaries). You can learn more about useful window functions in the corresponding vignette: `vignette("window-functions")`.

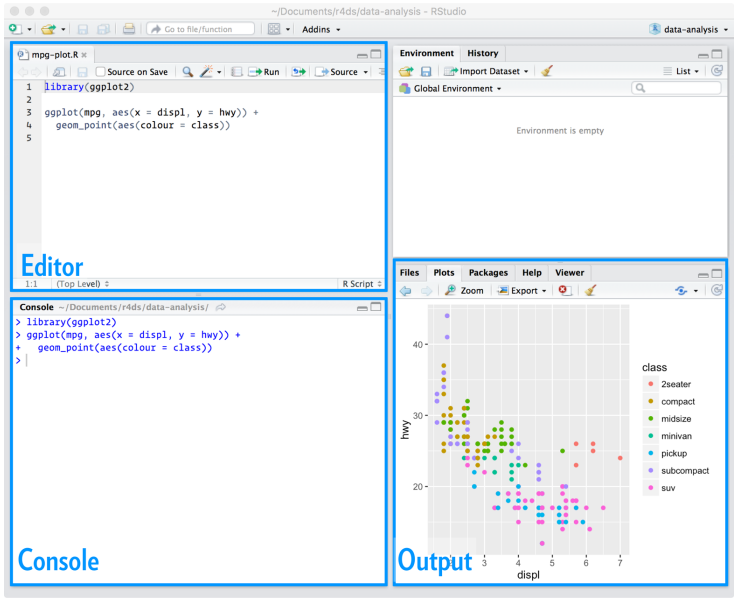
Exercises

1. Refer back to the table of useful mutate and filtering functions. Describe how each operation changes when you combine it with grouping.
2. Which plane (`tailnum`) has the worst on-time record?
3. What time of day should you fly if you want to avoid delays as much as possible?
4. For each destination, compute the total minutes of delay. For each flight, compute the proportion of the total delay for its destination.
5. Delays are typically temporally correlated: even once the problem that caused the initial delay has been resolved, later flights are delayed to allow earlier flights to leave. Using `lag()` explores how the delay of a flight is related to the delay of the immediately preceding flight.
6. Look at each destination. Can you find flights that are suspiciously fast? (That is, flights that represent a potential data entry error.) Compute the air time of a flight relative to the shortest flight to that destination. Which flights were most delayed in the air?

7. Find all destinations that are flown by at least two carriers. Use that information to rank the carriers.

CHAPTER 4
Workflow: Scripts

So far you've been using the console to run code. That's a great place to start, but you'll find it gets cramped pretty quickly as you create more complex **ggplot2** graphics and **dplyr** pipes. To give yourself more room to work, it's a great idea to use the script editor. Open it up either by clicking the File menu and selecting New File, then R script, or using the keyboard shortcut Cmd/Ctrl-Shift-N. Now you'll see four panes:



The script editor is a great place to put code you care about. Keep experimenting in the console, but once you have written code that works and does what you want, put it in the script editor. RStudio will automatically save the contents of the editor when you quit RStudio, and will automatically load it when you reopen. Nevertheless, it's a good idea to save your scripts regularly and to back them up.

Running Code

The script editor is also a great place to build up complex **ggplot2** plots or long sequences of **dplyr** manipulations. The key to using the script editor effectively is to memorize one of the most important keyboard shortcuts: Cmd/Ctrl-Enter. This executes the current R expression in the console. For example, take the following code. If your cursor is at █, pressing Cmd/Ctrl-Enter will run the complete command that generates `not_cancelled`. It will also move the cursor to the next statement (beginning with `not_cancelled %>%`). That makes it easy to run your complete script by repeatedly pressing Cmd/Ctrl-Enter:

```
library(dplyr)
library(nycflights13)

not_cancelled <- flights %>%
  filter(!is.na(dep_delay)█, !is.na(arr_delay))

not_cancelled %>%
  group_by(year, month, day) %>%
  summarize(mean = mean(dep_delay))
```

Instead of running expression-by-expression, you can also execute the complete script in one step: Cmd/Ctrl-Shift-S. Doing this regularly is a great way to check that you've captured all the important parts of your code in the script.

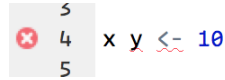
I recommend that you always start your script with the packages that you need. That way, if you share your code with others, they can easily see what packages they need to install. Note, however, that you should never include `install.packages()` or `setwd()` in a script that you share. It's very antisocial to change settings on someone else's computer!

When working through future chapters, I highly recommend starting in the editor and practicing your keyboard shortcuts. Over time,

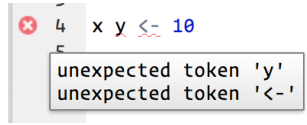
sending code to the console in this way will become so natural that you won't even think about it.

RStudio Diagnostics

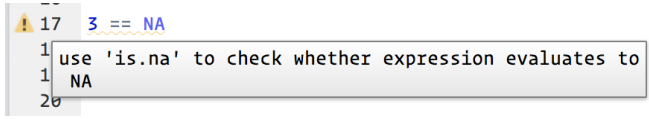
The script editor will also highlight syntax errors with a red squiggly line and a cross in the sidebar:



Hover over the cross to see what the problem is:



RStudio will also let you know about potential problems:



Exercises

1. Go to the RStudio Tips twitter account at *@rstudiotips* and find one tip that looks interesting. Practice using it!
2. What other common mistakes will RStudio diagnostics report? Read *http://bit.ly/RStudiocodediag* to find out.

CHAPTER 5

Exploratory Data Analysis

Introduction

This chapter will show you how to use visualization and transformation to explore your data in a systematic way, a task that statisticians call exploratory data analysis, or EDA for short. EDA is an iterative cycle. You:

1. Generate questions about your data.
2. Search for answers by visualizing, transforming, and modeling your data.
3. Use what you learn to refine your questions and/or generate new questions.

EDA is not a formal process with a strict set of rules. More than anything, EDA is a state of mind. During the initial phases of EDA you should feel free to investigate every idea that occurs to you. Some of these ideas will pan out, and some will be dead ends. As your exploration continues, you will hone in on a few particularly productive areas that you'll eventually write up and communicate to others.

EDA is an important part of any data analysis, even if the questions are handed to you on a platter, because you always need to investigate the quality of your data. Data cleaning is just one application of EDA: you ask questions about whether or not your data meets your expectations. To do data cleaning, you'll need to deploy all the tools of EDA: visualization, transformation, and modeling.

Prerequisites

In this chapter we'll combine what you've learned about **dplyr** and **ggplot2** to interactively ask questions, answer them with data, and then ask new questions.

```
library(tidyverse)
```

Questions

> There are no routine statistical questions, only questionable statistical routines.
>
> —Sir David Cox

> Far better an approximate answer to the right question, which is often vague, than an exact answer to the wrong question, which can always be made precise.
>
> —John Tukey

Your goal during EDA is to develop an understanding of your data. The easiest way to do this is to use questions as tools to guide your investigation. When you ask a question, the question focuses your attention on a specific part of your dataset and helps you decide which graphs, models, or transformations to make.

EDA is fundamentally a creative process. And like most creative processes, the key to asking *quality* questions is to generate a large *quantity* of questions. It is difficult to ask revealing questions at the start of your analysis because you do not know what insights are contained in your dataset. On the other hand, each new question that you ask will expose you to a new aspect of your data and increase your chance of making a discovery. You can quickly drill down into the most interesting parts of your data—and develop a set of thought-provoking questions—if you follow up each question with a new question based on what you find.

There is no rule about which questions you should ask to guide your research. However, two types of questions will always be useful for making discoveries within your data. You can loosely word these questions as:

1. What type of variation occurs within my variables?
2. What type of covariation occurs between my variables?

The rest of this chapter will look at these two questions. I'll explain what variation and covariation are, and I'll show you several ways to answer each question. To make the discussion easier, let's define some terms:

- A *variable* is a quantity, quality, or property that you can measure.
- A *value* is the state of a variable when you measure it. The value of a variable may change from measurement to measurement.
- An *observation*, or a *case*, is a set of measurements made under similar conditions (you usually make all of the measurements in an observation at the same time and on the same object). An observation will contain several values, each associated with a different variable. I'll sometimes refer to an observation as a data point.
- *Tabular data* is a set of values, each associated with a variable and an observation. Tabular data is *tidy* if each value is placed in its own "cell," each variable in its own column, and each observation in its own row.

So far, all of the data that you've seen has been tidy. In real life, most data isn't tidy, so we'll come back to these ideas again in Chapter 9.

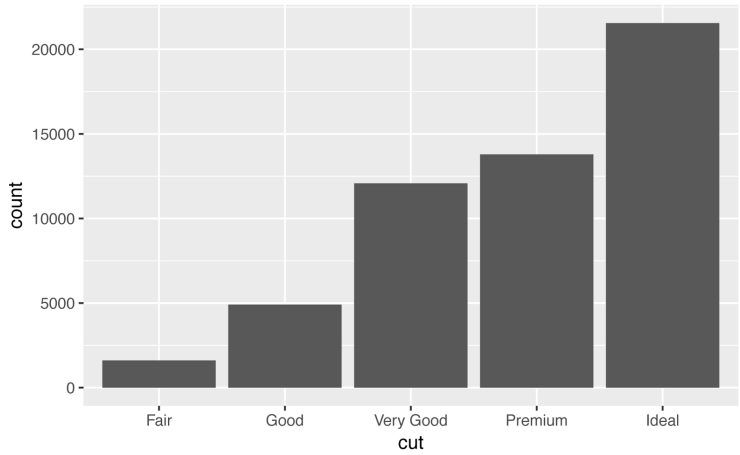
Variation

Variation is the tendency of the values of a variable to change from measurement to measurement. You can see variation easily in real life; if you measure any continuous variable twice, you will get two different results. This is true even if you measure quantities that are constant, like the speed of light. Each of your measurements will include a small amount of error that varies from measurement to measurement. Categorical variables can also vary if you measure across different subjects (e.g., the eye colors of different people), or different times (e.g., the energy levels of an electron at different moments). Every variable has its own pattern of variation, which can reveal interesting information. The best way to understand that pattern is to visualize the distribution of variables' values.

Visualizing Distributions

How you visualize the distribution of a variable will depend on whether the variable is categorical or continuous. A variable is *categorical* if it can only take one of a small set of values. In R, categorical variables are usually saved as factors or character vectors. To examine the distribution of a categorical variable, use a bar chart:

```
ggplot(data = diamonds) +
  geom_bar(mapping = aes(x = cut))
```

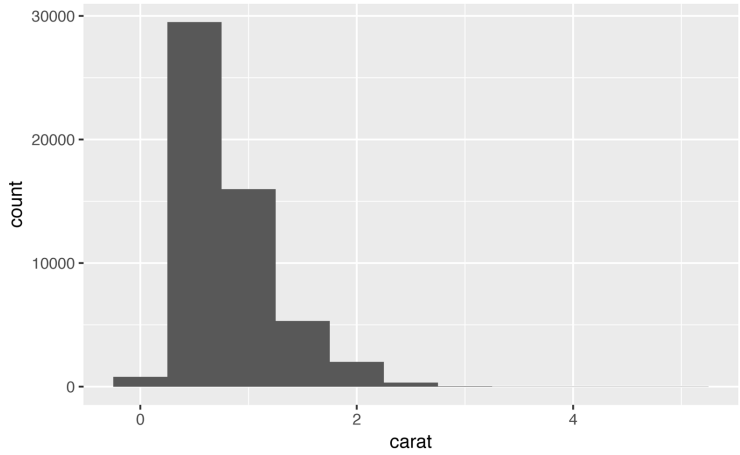


The height of the bars displays how many observations occurred with each x value. You can compute these values manually with `dplyr::count()`:

```
diamonds %>%
  count(cut)
#> # A tibble: 5 × 2
#>         cut     n
#>       <ord> <int>
#> 1      Fair  1610
#> 2      Good  4906
#> 3 Very Good 12082
#> 4   Premium 13791
#> 5     Ideal 21551
```

A variable is *continuous* if it can take any of an infinite set of ordered values. Numbers and date-times are two examples of continuous variables. To examine the distribution of a continuous variable, use a histogram:

```
ggplot(data = diamonds) +
  geom_histogram(mapping = aes(x = carat), binwidth = 0.5)
```



You can compute this by hand by combining `dplyr::count()` and `ggplot2::cut_width()`:

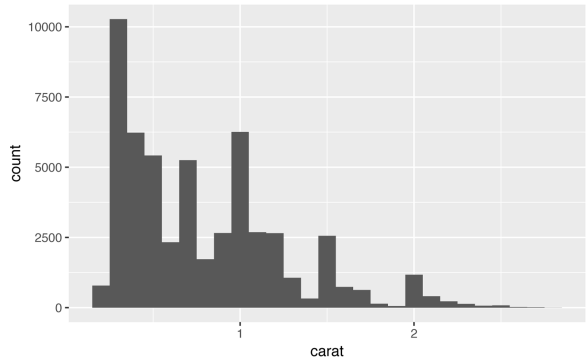
```
diamonds %>%
  count(cut_width(carat, 0.5))
#> # A tibble: 11 × 2
#>   `cut_width(carat, 0.5)`     n
#>                    <fctr> <int>
#> 1           [-0.25,0.25]   785
#> 2            (0.25,0.75] 29498
#> 3            (0.75,1.25] 15977
#> 4            (1.25,1.75]  5313
#> 5            (1.75,2.25]  2002
#> 6            (2.25,2.75]   322
#> # ... with 5 more rows
```

A histogram divides the x-axis into equally spaced bins and then uses the height of each bar to display the number of observations that fall in each bin. In the preceding graph, the tallest bar shows that almost 30,000 observations have a `carat` value between 0.25 and 0.75, which are the left and right edges of the bar.

You can set the width of the intervals in a histogram with the `binwidth` argument, which is measured in the units of the `x` variable. You should always explore a variety of binwidths when working with histograms, as different binwidths can reveal different patterns. For example, here is how the preceding graph looks when we zoom

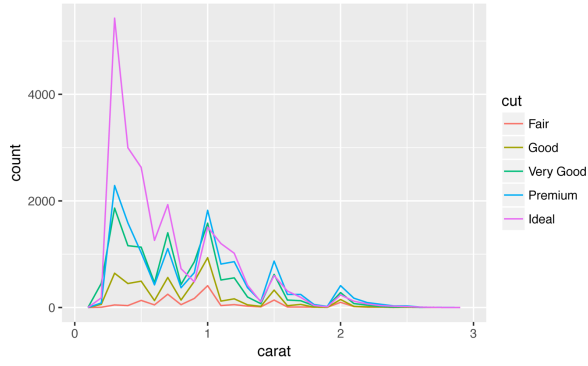
into just the diamonds with a size of less than three carats and choose a smaller binwidth:

```
smaller <- diamonds %>%
  filter(carat < 3)

ggplot(data = smaller, mapping = aes(x = carat)) +
  geom_histogram(binwidth = 0.1)
```



If you wish to overlay multiple histograms in the same plot, I recommend using `geom_freqpoly()` instead of `geom_histogram()`. `geom_freqpoly()` performs the same calculation as `geom_histogram()`, but instead of displaying the counts with bars, uses lines instead. It's much easier to understand overlapping lines than bars:

```
ggplot(data = smaller, mapping = aes(x = carat, color = cut)) +
  geom_freqpoly(binwidth = 0.1)
```



There are a few challenges with this type of plot, which we will come back to in "A Categorical and Continuous Variable" on page 93.

Typical Values

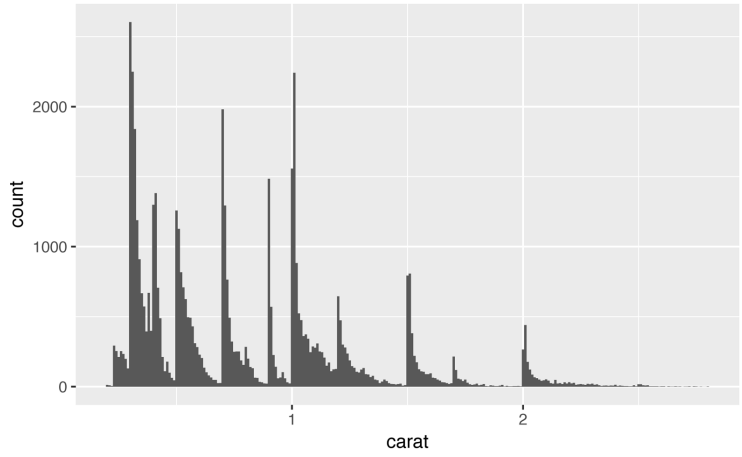
In both bar charts and histograms, tall bars show the common values of a variable, and shorter bars show less-common values. Places that do not have bars reveal values that were not seen in your data. To turn this information into useful questions, look for anything unexpected:

- Which values are the most common? Why?
- Which values are rare? Why? Does that match your expectations?
- Can you see any unusual patterns? What might explain them?

As an example, the following histogram suggests several interesting questions:

- Why are there more diamonds at whole carats and common fractions of carats?
- Why are there more diamonds slightly to the right of each peak than there are slightly to the left of each peak?
- Why are there no diamonds bigger than 3 carats?

```
ggplot(data = smaller, mapping = aes(x = carat)) +
  geom_histogram(binwidth = 0.01)
```

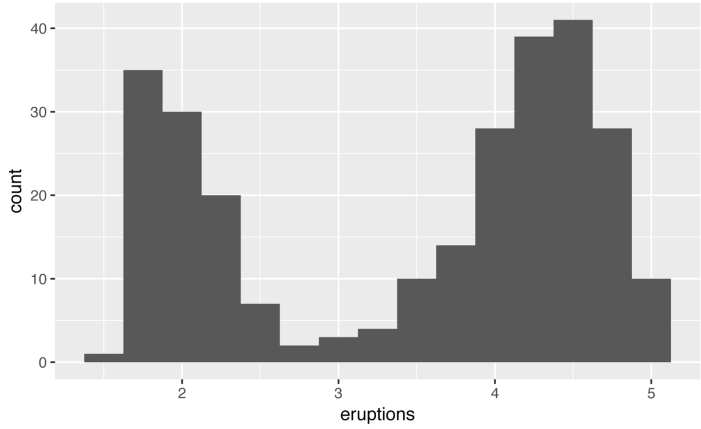


In general, clusters of similar values suggest that subgroups exist in your data. To understand the subgroups, ask:

- How are the observations within each cluster similar to each other?
- How are the observations in separate clusters different from each other?
- How can you explain or describe the clusters?
- Why might the appearance of clusters be misleading?

The following histogram shows the length (in minutes) of 272 eruptions of the Old Faithful Geyser in Yellowstone National Park. Eruption times appear to be clustered into two groups: there are short eruptions (of around 2 minutes) and long eruptions (4–5 minutes), but little in between:

```
ggplot(data = faithful, mapping = aes(x = eruptions)) +
  geom_histogram(binwidth = 0.25)
```



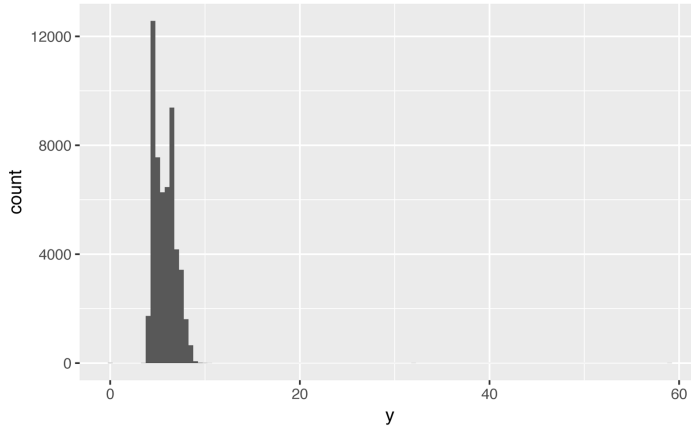
Many of the preceding questions will prompt you to explore a relationship *between* variables, for example, to see if the values of one variable can explain the behavior of another variable. We'll get to that shortly.

Unusual Values

Outliers are observations that are unusual; data points that don't seem to fit the pattern. Sometimes outliers are data entry errors; other times outliers suggest important new science. When you have a lot of data, outliers are sometimes difficult to see in a histogram. For example, take the distribution of the `y` variable from the dia-

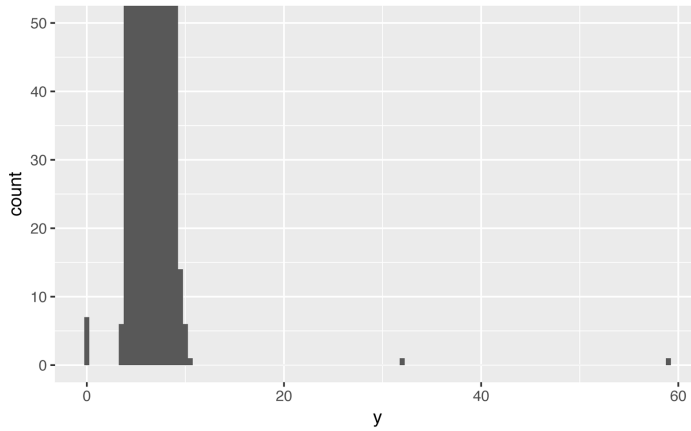
monds dataset. The only evidence of outliers is the unusually wide limits on the y-axis:

```
ggplot(diamonds) +
  geom_histogram(mapping = aes(x = y), binwidth = 0.5)
```



There are so many observations in the common bins that the rare bins are so short that you can't see them (although maybe if you stare intently at 0 you'll spot something). To make it easy to see the unusual values, we need to zoom in to small values of the y-axis with `coord_cartesian()`:

```
ggplot(diamonds) +
  geom_histogram(mapping = aes(x = y), binwidth = 0.5) +
  coord_cartesian(ylim = c(0, 50))
```



(`coord_cartesian()` also has an `xlim()` argument for when you need to zoom into the x-axis. **ggplot2** also has `xlim()` and `ylim()` functions that work slightly differently: they throw away the data outside the limits.)

This allows us to see that there are three unusual values: 0, ~30, and ~60. We pluck them out with **dplyr**:

```
unusual <- diamonds %>%
  filter(y < 3 | y > 20) %>%
  arrange(y)
unusual
#> # A tibble: 9 × 10
#>   carat       cut color clarity depth table price     x
#>   <dbl>     <ord> <ord>   <ord> <dbl> <dbl> <int> <dbl>
#> 1  1.00 Very Good     H     VS2  63.3    53  5139  0.00
#> 2  1.14      Fair     G     VS1  57.5    67  6381  0.00
#> 3  1.56     Ideal     G     VS2  62.2    54 12800  0.00
#> 4  1.20   Premium     D    VVS1  62.1    59 15686  0.00
#> 5  2.25   Premium     H     SI2  62.8    59 18034  0.00
#> 6  0.71      Good     F     SI2  64.1    60  2130  0.00
#> 7  0.71      Good     F     SI2  64.1    60  2130  0.00
#> 8  0.51     Ideal     E     VS1  61.8    55  2075  5.15
#> 9  2.00   Premium     H     SI2  58.9    57 12210  8.09
#> # ... with 2 more variables:
#> #   y <dbl>, z <dbl>
```

The `y` variable measures one of the three dimensions of these diamonds, in mm. We know that diamonds can't have a width of 0mm, so these values must be incorrect. We might also suspect that measurements of 32mm and 59mm are implausible: those diamonds are over an inch long, but don't cost hundreds of thousands of dollars!

It's good practice to repeat your analysis with and without the outliers. If they have minimal effect on the results, and you can't figure out why they're there, it's reasonable to replace them with missing values and move on. However, if they have a substantial effect on your results, you shouldn't drop them without justification. You'll need to figure out what caused them (e.g., a data entry error) and disclose that you removed them in your write-up.

Exercises

1. Explore the distribution of each of the `x`, `y`, and `z` variables in `diamonds`. What do you learn? Think about a diamond and how you might decide which dimension is the length, width, and depth.

2. Explore the distribution of `price`. Do you discover anything unusual or surprising? (Hint: carefully think about the `bin width` and make sure you try a wide range of values.)
3. How many diamonds are 0.99 carat? How many are 1 carat? What do you think is the cause of the difference?
4. Compare and contrast `coord_cartesian()` versus `xlim()` or `ylim()` when zooming in on a histogram. What happens if you leave `binwidth` unset? What happens if you try and zoom so only half a bar shows?

Missing Values

If you've encountered unusual values in your dataset, and simply want to move on to the rest of your analysis, you have two options:

- Drop the entire row with the strange values:

```
diamonds2 <- diamonds %>%
  filter(between(y, 3, 20))
```

  I don't recommend this option as just because one measurement is invalid, doesn't mean all the measurements are. Additionally, if you have low-quality data, by time that you've applied this approach to every variable you might find that you don't have any data left!

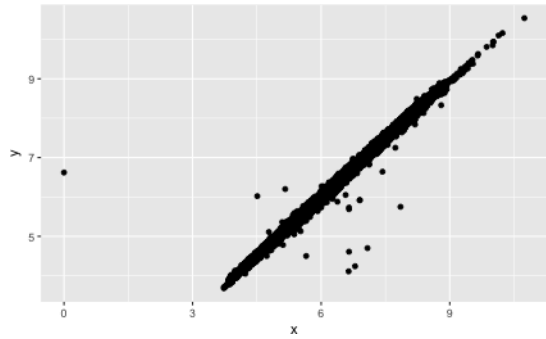
- Instead, I recommend replacing the unusual values with missing values. The easiest way to do this is to use `mutate()` to replace the variable with a modified copy. You can use the `ifelse()` function to replace unusual values with `NA`:

```
diamonds2 <- diamonds %>%
  mutate(y = ifelse(y < 3 | y > 20, NA, y))
```

`ifelse()` has three arguments. The first argument `test` should be a logical vector. The result will contain the value of the second argument, `yes`, when `test` is `TRUE`, and the value of the third argument, `no`, when it is false.

Like R, **ggplot2** subscribes to the philosophy that missing values should never silently go missing. It's not obvious where you should plot missing values, so **ggplot2** doesn't include them in the plot, but it does warn that they've been removed:

```
ggplot(data = diamonds2, mapping = aes(x = x, y = y)) +
  geom_point()
#> Warning: Removed 9 rows containing missing values
#> (geom_point).
```

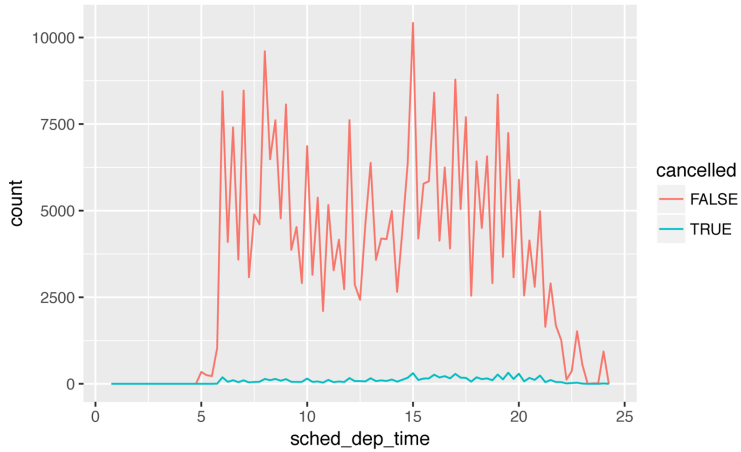


To suppress that warning, set `na.rm = TRUE`:

```
ggplot(data = diamonds2, mapping = aes(x = x, y = y)) +
  geom_point(na.rm = TRUE)
```

Other times you want to understand what makes observations with missing values different from observations with recorded values. For example, in `nycflights13::flights`, missing values in the `dep_time` variable indicate that the flight was cancelled. So you might want to compare the scheduled departure times for cancelled and noncancelled times. You can do this by making a new variable with `is.na()`:

```
nycflights13::flights %>%
  mutate(
    cancelled = is.na(dep_time),
    sched_hour = sched_dep_time %/% 100,
    sched_min = sched_dep_time %% 100,
    sched_dep_time = sched_hour + sched_min / 60
  ) %>%
  ggplot(mapping = aes(sched_dep_time)) +
    geom_freqpoly(
      mapping = aes(color = cancelled),
      binwidth = 1/4
    )
```



However, this plot isn't great because there are many more non-cancelled flights than cancelled flights. In the next section we'll explore some techniques for improving this comparison.

Exercises

1. What happens to missing values in a histogram? What happens to missing values in a bar chart? Why is there a difference?
2. What does `na.rm = TRUE` do in `mean()` and `sum()`?

Covariation

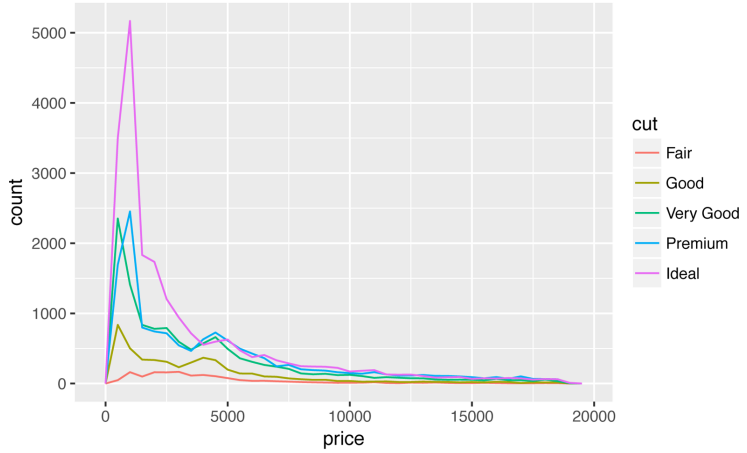
If variation describes the behavior *within* a variable, covariation describes the behavior *between* variables. *Covariation* is the tendency for the values of two or more variables to vary together in a related way. The best way to spot covariation is to visualize the relationship between two or more variables. How you do that should again depend on the type of variables involved.

A Categorical and Continuous Variable

It's common to want to explore the distribution of a continuous variable broken down by a categorical variable, as in the previous frequency polygon. The default appearance of `geom_freqpoly()` is not that useful for that sort of comparison because the height is

given by the count. That means if one of the groups is much smaller than the others, it's hard to see the differences in shape. For example, let's explore how the price of a diamond varies with its quality:

```
ggplot(data = diamonds, mapping = aes(x = price)) +
  geom_freqpoly(mapping = aes(color = cut), binwidth = 500)
```



It's hard to see the difference in distribution because the overall counts differ so much:

```
ggplot(diamonds) +
  geom_bar(mapping = aes(x = cut))
```